



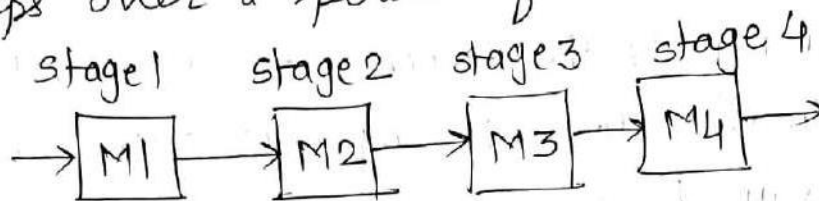
DEPARTMENT OF COMPUTER ENGINEERING

PIPELINING

- Pipelining is the process of arrangement of hardware elements of CPU such that its overall performance is increased.
- The pipeline design provides a way to start a new task before an old one has been completed.
- The concept can be understood through an example of industrial plant.
- To achieve pipelining one must ^{sub}divide the ip task into a sequence of subtasks.
- Each subtasks can be processed independently on a different machine.
- Subtasks of several tasks can be processed concurrently.
- Successive tasks are streamed into the pipe and get executed in an overlapped fashion at the subtask level.



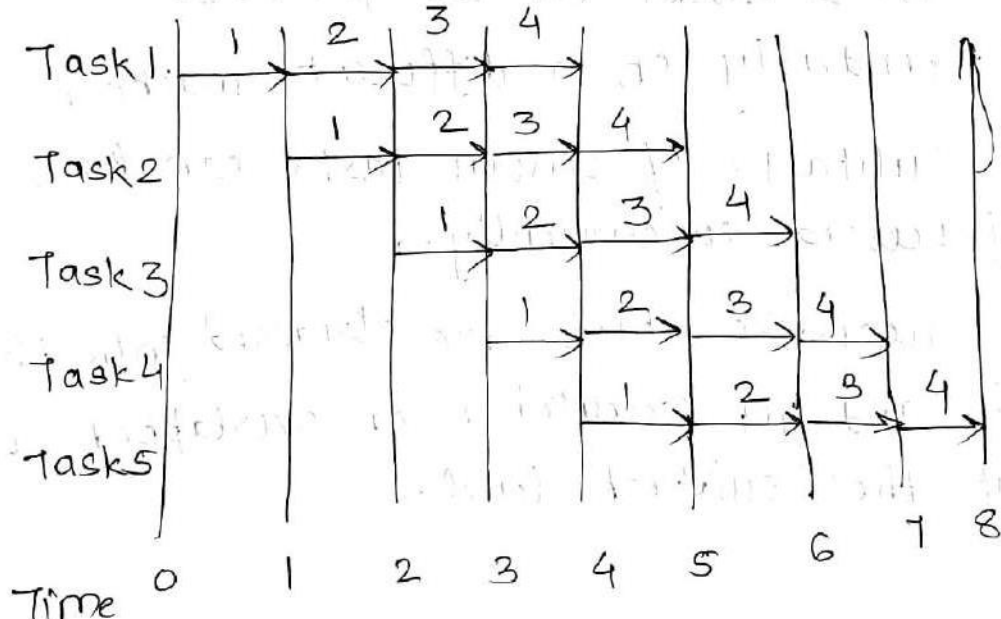
→ In the fig below, a sequential process is being done step by step in 4 steps over a period of time.



→ In order to complete a task it must be processed sequentially in machine M1, M2, M3 and M4.

→ After processing of first task is done on machine M1, it moves to machine M2.

→ Now machine M1 is free and 2nd task can be loaded on machine M1.





→ Task 1 is completed in 4 time intervals

→ During the time interval 3-4

task 1 is being processed on M/c M4

" 2 " " " " M/c M3

" 3 " " " " M/c M2

" 4 " " " " M/c M1.

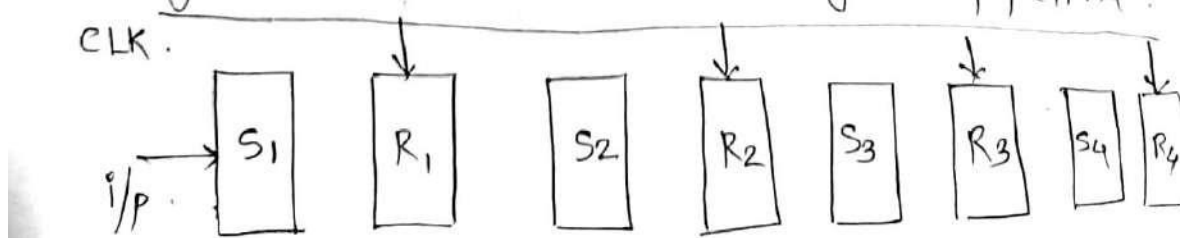
→ Thus during any time interval upto 4 jobs could be getting processed concurrently.

→ Though it may appear that it should be possible to execute any applicn in various pipeline stages, it is not actually so.

→ Only those applications which can be broken down into independent subtasks may take advantage of pipelining.

→ It is most efficient for those applicn that need to repeat the same task many times with different sets of data.

→ Fig below shows the 4 segment pipeline.





- The fig shows 2 kinds of modules S_i and R_i
- These specify the combinational ckt and the registers to hold the results resp.
- The information flows from left to right.
- The behaviour of a pipelined operation can be explained with the help of a space time diagram as shown below
- The horizontal axis displays the time in clk cycle and vertical axis displays the task number.

	1	2	3	4	5	6	7	8	9	10
T_1	S_1	S_2	S_3	S_4						
T_2		S_1	S_2	S_3	S_4					
T_3			S_1	S_2	S_3	S_4				
T_4				S_1	S_2	S_3	S_4			
T_5					S_1	S_2	S_3	S_4		



- The diagram shows 5 tasks T_1 to T_5 .
- A task is completed after all its subtasks S_1 to S_4 are completed.
- In initial clk cycle : segment 1 $\leftarrow$ Task 1.
while all other segments are idle.
- In second clk cycle segment 1 $\leftarrow$ Task 2
- Once the pipe is filled up, it will o/p ^{Segment} one result per clk period, independent of the no. of stages in the pipe.
- let us consider the case of k stage pipeline.
- let us clk ~~per~~ cycle time is T_p and total n tasks are to be executed.
- Time reqd to complete first Task = $k T_p$
- the remaining $(n-1)$ tasks emerge from the pipe at the rate of 1 task per ^{clk} cycle and they will be completed after a period equal to $(n-1) T_p$.
- ∴ Time reqd to complete n tasks
= $k + (n-1)$ clock periods.



Advantages of Pipelining

1. The cycle time of the processor is reduced.
2. It increases the throughput of the system
3. It makes the system reliable.

Disadvantages of Pipelining

1. The design of pipelined processor is complex and costly to manufacture.
2. The instruction latency is more.

Pipeline Stages

Following are the 5 stages of pipeline with their respective operations:

- **Stage 1 (Instruction Fetch)**
In this stage the CPU reads instructions from the address in the memory whose value is present in the program counter.
- **Stage 2 (Instruction Decode)**
In this stage, instruction is decoded and the register file is accessed to get the values from the registers used in the instruction.
- **Stage 3 (Instruction Execute)**
In this stage, ALU operations are performed.
- **Stage 4 (Memory Access)**
In this stage, memory operands are read and written from/to the memory that is present in the instruction.
- **Stage 5 (Write Back)**
In this stage, computed/fetched value is written back to the register present in the instructions.

Performance of Pipelined Execution-

The following parameters serve as criterion to estimate

Speed Up

Efficiency

Throughput



1. Speed Up-

It gives an idea of “**how much faster**” the pipelined execution is as compared to non-pipelined execution.

It is calculated as-

$$\text{Speed Up (S)} = \frac{\text{Non-pipelined execution time}}{\text{Pipelined execution time}}$$

2. Efficiency-

The efficiency

$$\text{Efficiency } (\eta) = \frac{\text{Speed Up}}{\text{Number of stages in Pipelined Architecture}}$$

OR

$$\text{Efficiency } (\eta) = \frac{\text{Number of boxes utilized in phase-time diagram}}{\text{Total number of boxes in phase-time diagram}}$$

3. Throughput-

Throughput is defined as number of instructions executed per unit time.

It is calculated as-

$$\text{Throughput} = \frac{\text{Number of instructions executed}}{\text{Total time taken}}$$



Response time is the time from start to completion of a task. This also includes:

- Operating system overhead.
- Waiting for I/O and other processes
- Accessing disk and memory
- Time spent executing on the CPU or execution time.

Throughput is the total amount of work done in a given time.

CPU execution time is the total time a CPU spends computing on a given task. It also excludes time for I/O or running other programs. This is also referred to as simply CPU time.

Performance is determined by execution time as performance is inversely proportional to execution time.

How to Improve Performance?

To improve performance you can either:

- Decrease the CPI (clock cycles per instruction) by using new Hardware.
- Decrease the clock time or Increase clock rate by reducing propagation delays or by use pipelining.
- Decrease the number of required cycles or improve ISA or Compiler.

In instruction pipelining,

- A form of parallelism called as instruction level parallelism is implemented.
- Multiple instructions execute simultaneously.
- The efficiency of pipelined execution is more than that of non-pipelined execution.